Quantification: 701_z26c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	14
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1922.92	1500	1365.51	px^2
major_axis_length_px	lacuna shape	75.4752	67.0082	29.7312	px
minor_axis_length_px	lacuna shape	38.0255	32.9459	15.5667	px
aspect_ratio	lacuna shape	2.1407	1.9822	0.8206	unitless
eccentricity	lacuna shape	0.7964	0.8634	0.1957	unitless
solidity	lacuna shape	0.7414	0.7609	0.0869	unitless
roots_count	headline	10.2143	8	4.7423	count
ring_length_r30_px	headline	431.929	434	162.838	px
ring_length_r60_px	reported	997.857	1004.5	399.938	px
owned_length_px	ownership-dependent	1637.67	1176.07	1332.31	px
edge_count	ownership-dependent	111.071	77	91.9653	count
mean_edge_length_px	ownership-dependent	14.9048	14.776	1.7904	px
perimeter_px	normalised	278.299	209.025	156.449	px
ring_area_r30_px2	normalised	8450.57	7637	2918.54	px^2
ring_area_r60_px2	normalised	20263.6	19812	5896.35	px^2
in_frame_fraction_r30	normalised	0.9961	1	0.0145	unitless
in_frame_fraction_r60	normalised	0.9868	1	0.0389	unitless
ring_density_r30	normalised	0.0512776	0.0496469	0.0105077	px^-1
ring_density_r60	normalised	0.0489872	0.0475719	0.0135207	px^-1
roots_per_100px_perimeter	normalised	3.9184	4.0381	0.7987	per 100 px
ring_attached_length_r30_px	network v2	364.214	335	174.115	px
ring_attached_length_r60_px	network v2	699.786	765.5	426.968	px
ring_length_w_r30_px	network v2	489.421	487.454	183.865	px
ring_length_w_r60_px	network v2	1124.69	1137.05	449.617	px
sholl_crossings_r10	network v2	10.3571	8.5	5.1681	count
sholl_crossings_r20	network v2	11.9286	11	4.3759	count
sholl_crossings_r30	network v2	12.7857	12	5.5633	count
loop_count	loops, report only	8.1429	2	10.6688	count
loop_edge_share	loops, report only	0.1822	0.1118	0.1885	share
ring_width_mean_r30_px	width	6.3005	6.1681	0.79	px

Per field
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measure	value	units
analysed_area_px2	762050	px^2
total_skeleton_length_px	37632	px
canalicular_length_density_per_px	0.0493826	px^-1
junction_count	1077	count
junction_density_per_px2	0.00141329	px^-2
lacuna_count	14	count

lacunae_per_px2	1.837e-05	px^-2
skeleton_component_count	661	count
mean_component_length_px	56.9319	px
median_component_length_px	13	px
field_length_density_w_per_px	0.0553969	px^-1
field_density_without_flagged_per_px	0.0340055	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	226	px
unattached_component_count	15	count
canalicular_length_density_attached_per_px	0.0493826	px^-1
field_length_density_with_unattached_per_px	0.0496791	px^-1
excluded_area_px2	261257	px^2
analysed_field_area_px2	787319	px^2
loop_count	189	count
loop_edge_share	0.2153	share
edge_count	3539	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.3721	px
width_median_px	6	px
width_p10_px	2.8284	px
width_p90_px	10	px
width_pixels_used	29836	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2287	125.157	32.9459	3.7989	0.9647	0.6065	0.5498	75.4101	463.222	False
2	5110	133.797	67.4988	1.9822	0.8634	0.6757	1.2061	267.96	890.703	False
3	1782	62.6142	47.9495	1.3058	0.6431	0.682	1.1297	243.333	206.608	False
4	1422	68.8124	28.5123	2.4134	0.9101	0.8335	1.1259	314.225	198.443	False
5	1652	67.0082	31.8663	2.1028	0.8797	0.8959	0.9049	359.245	256.938	False
6	967	67.0717	21.9089	3.0614	0.9451	0.7662	-1.3191	319.159	644.63	False
7	821	63.8935	20.0569	3.1856	0.9495	0.7609	0.1039	594.211	741.577	False
8	2272	79.8087	42.6154	1.8728	0.8455	0.7881	-1.0855	686.797	269.451	False
9	656	33.1626	30.2759	1.0953	0.4081	0.801	1.2644	753.977	863.463	False
10	1435	61.7195	33.8344	1.8242	0.8363	0.8112	-1.4959	787.26	645.571	False
11	4376	111.203	63.8724	1.741	0.8186	0.6554	-1.0966	845.546	482.61	False
12	1500	55.4417	51.2017	1.0828	0.3835	0.6268	-1.2338	890.901	936.497	False
13	718	51.4892	21.7937	2.3626	0.906	0.7349	-1.184	917.613	557.102	False
14								904.956	141.983	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	20	578	1268	873.352	66	13.2326	6.7073
2	16	607	1418	2223.15	146	15.2271	6.3554
3	12	431	740	346.007	26	13.308	5.3499

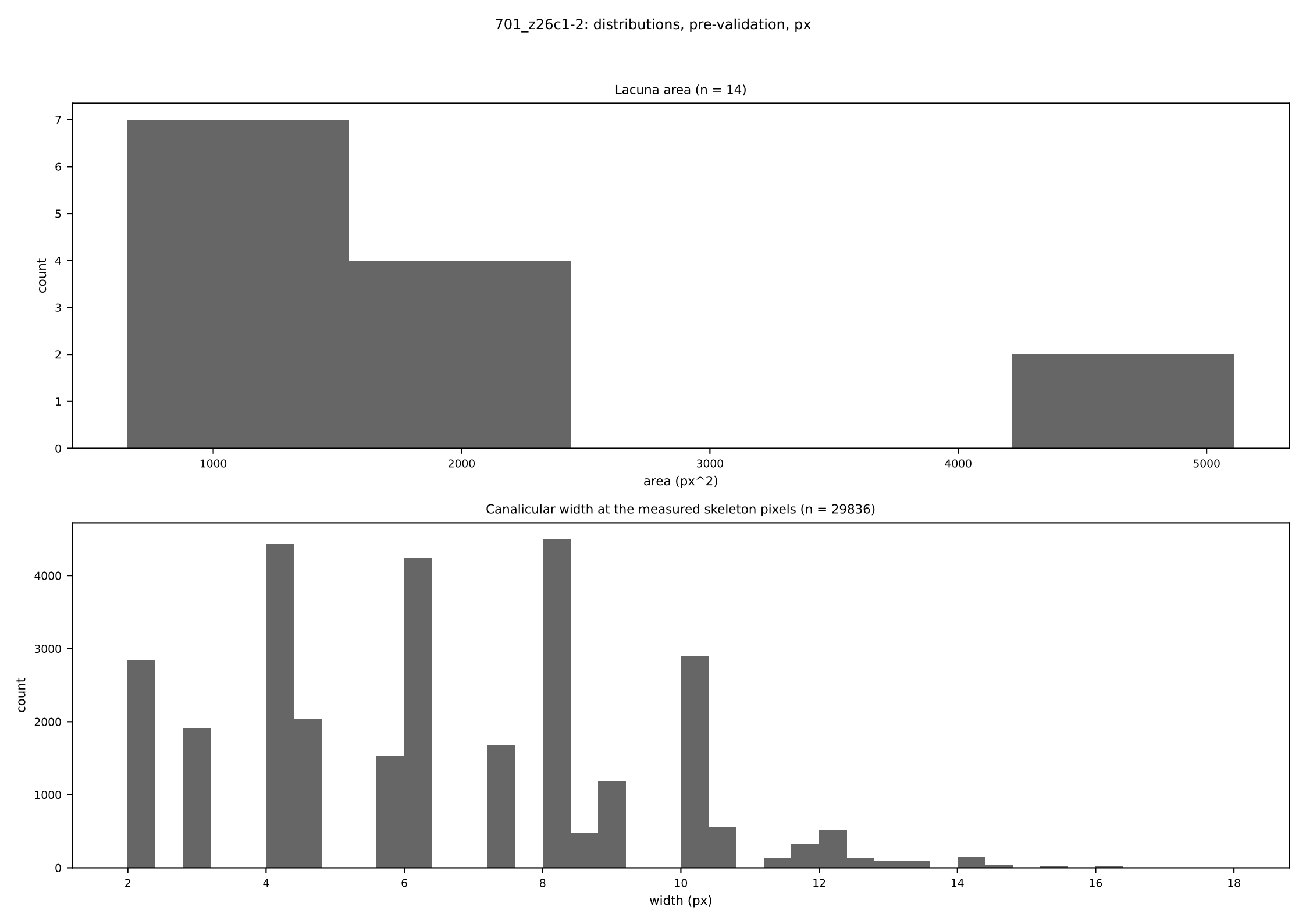
4	7	233	386	426.274	27	15.7879	5.8741
5	6	264	714	465.274	23	20.2293	5.9808
6	8	303	689	478.576	34	14.0758	6.3791
7	7	360	922	1248.39	80	15.6049	6.6728
8	8	437	1148	2672.81	177	15.1006	5.9552
9	7	259	461	303.208	24	12.6337	5.4386
10	10	510	1151	2313.98	159	14.5534	6.8738
11	18	805	1914	4408.86	310	14.2221	5.6861
12	12	478	1077	1103.75	74	14.9155	5.6599
13	7	497	1150	2331.43	154	15.1391	8.3653
14	5	285	932	3732.34	255	14.6366	6.9089

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	459.683	11984	25966	0.9457	0.8575	0.048231	0.0488331	4.3508
2	541.524	14453	33199	1	1	0.0419982	0.0427121	2.9546
3	345.031	7948	17293	1	1	0.0542275	0.0427919	3.4779
4	202.35	5401	10020	1	1	0.0431402	0.038523	3.4593
5	196.522	6388	15414	1	1	0.0413275	0.0463215	3.0531
6	188.244	7408	20330	1	1	0.0409017	0.0338908	4.2498
7	163.723	6981	19532	1	1	0.0515685	0.0472046	4.2755
8	273.019	9285	23947	1	1	0.0470652	0.0479392	2.9302
9	133.338	6168	17509	1	1	0.0419909	0.0263293	5.2498
10	215.7	7866	20092	1	1	0.064836	0.0572865	4.6361
11	605.222	13350	26836	1	1	0.0602996	0.0713221	2.9741
12	307.298	9361	22441	1	0.9574	0.0510629	0.0479925	3.905
13	167.823	6421	14897	1	1	0.0774023	0.0771967	4.1711
14	96.7107	5294	16215	1	1	0.0538345	0.0574776	5.1701

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	529	861	659.985	1432.43	22	10	12
2	582	1006	692.754	1603.69	18	20	18
3	238	266	475.777	818.683	9	11	11
4	162	206	266.693	442.688	7	6	6
5	146	182	303.291	823.394	6	11	7
6	215	344	335.936	766.512	6	7	9
7	297	496	417.333	1041.55	7	10	16
8	373	853	499.132	1289.22	8	14	14
9	220	284	295.522	508.889	5	8	4
10	480	1028	575.375	1291.86	11	14	18
11	713	1660	906.482	2143.63	17	20	24
12	410	729	540.688	1215.67	11	13	18
13	490	1080	563.132	1309.05	11	15	12
14	244	802	319.794	1058.44	7	8	10


701_z26c1-2: distributions, pre-validation, px
Lacuna area (n = 14)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 29836)
count
width (px)